

Lads vs Nups

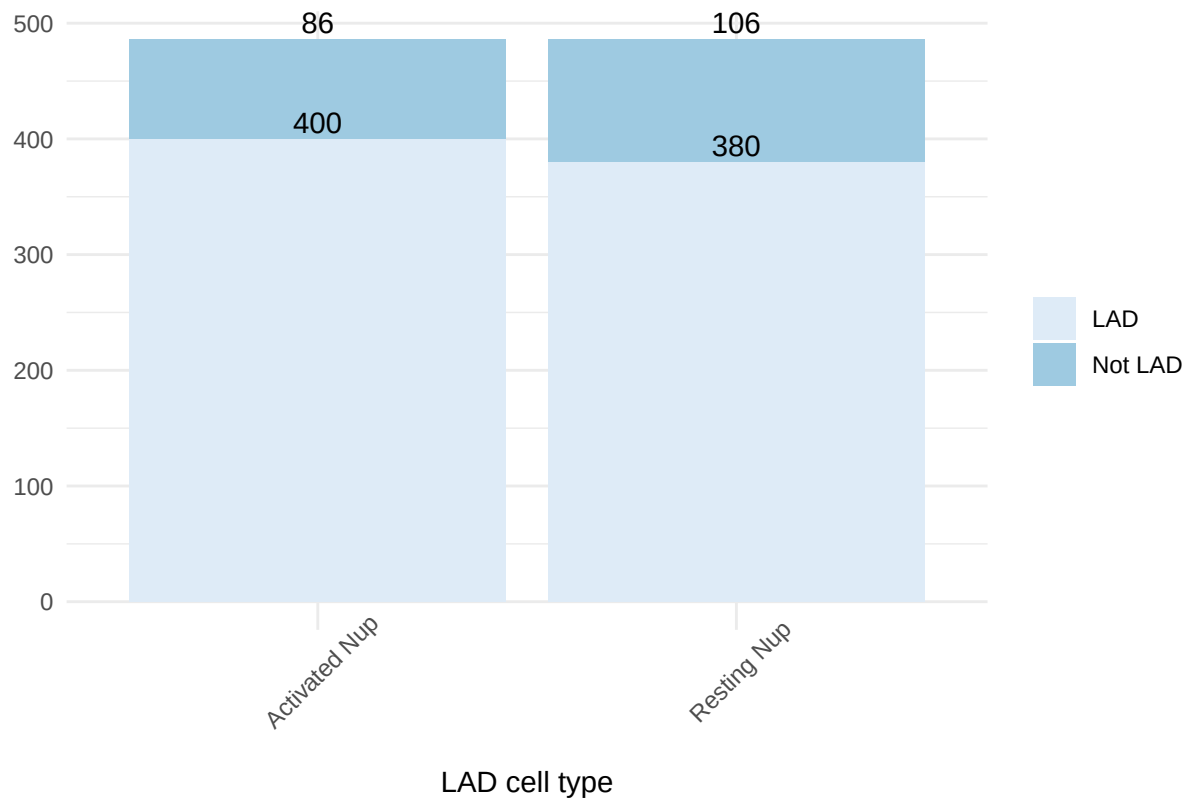
Comparison of Nup related genes and Robson LADs / Belmont SPIN classification

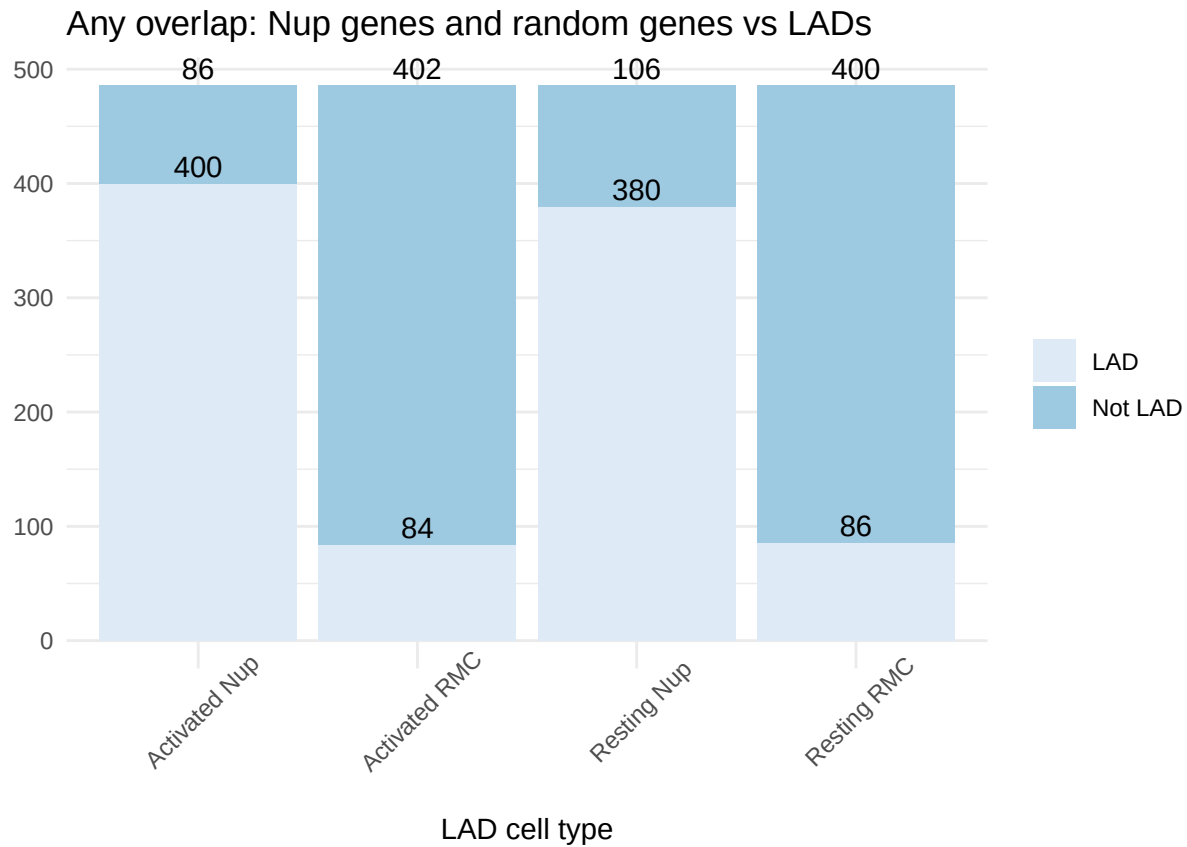
Robson LADs first

1. Overlap of Nup related genes and LADs

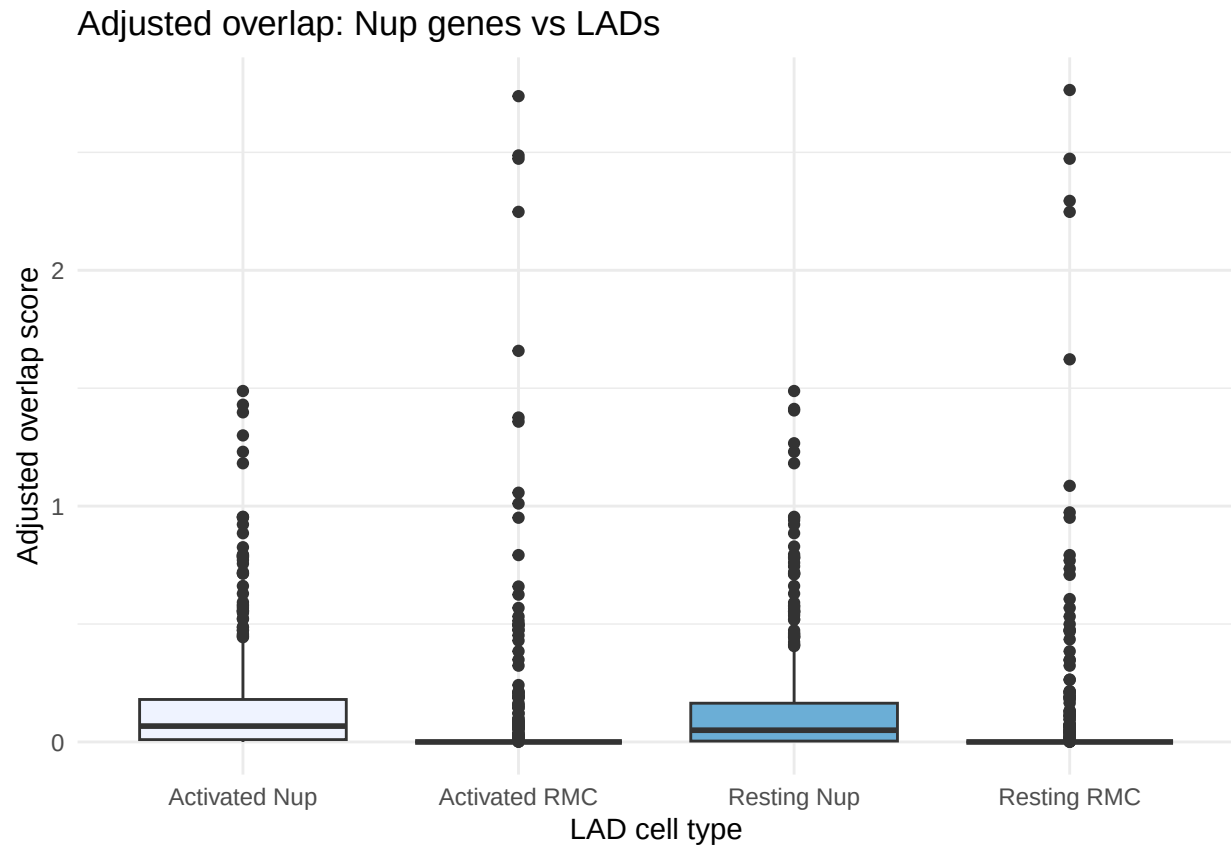
If there is any overlap between Nup gene and LAD the data point is treated as LAD. RMCs are randomly chosen protein coding genes with sample size == 486.

Any overlap: Nup genes vs LADs



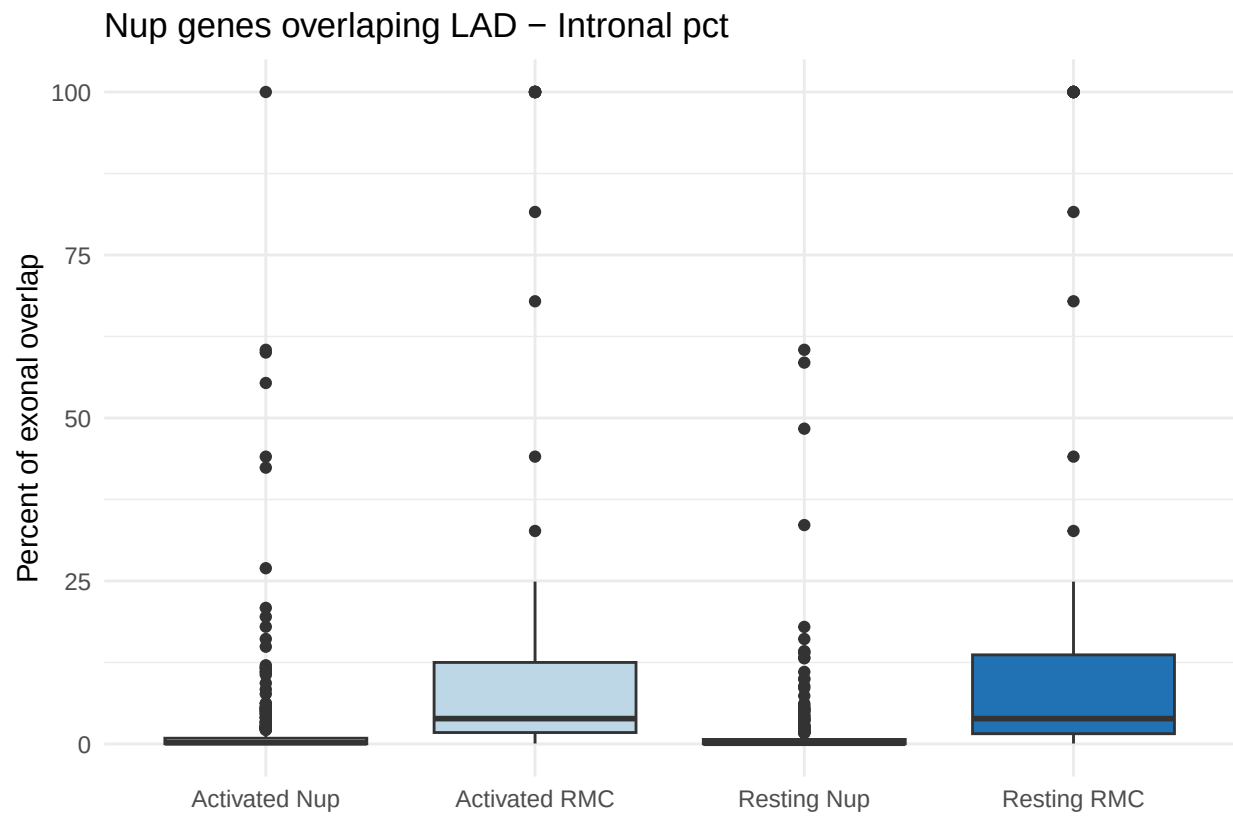


Problem with above graphs: Nup genes are around 2x wider than an average random selection of protein coding genes. This can't explain the full difference but here's the same data but weighted by gene width. Adjusted overlap score is the number of bp that overlap LAD divided by total length of all genes in group (Nups or RMCs). The difference is less extreme but still statistically significant.



2. Overview of the interaction between intron/exon split in Nup genes and Robson LADs.

Percentatge overlap thats in exons for genes that have any overlap at all; the Nup gene overlap is mostly intronal:



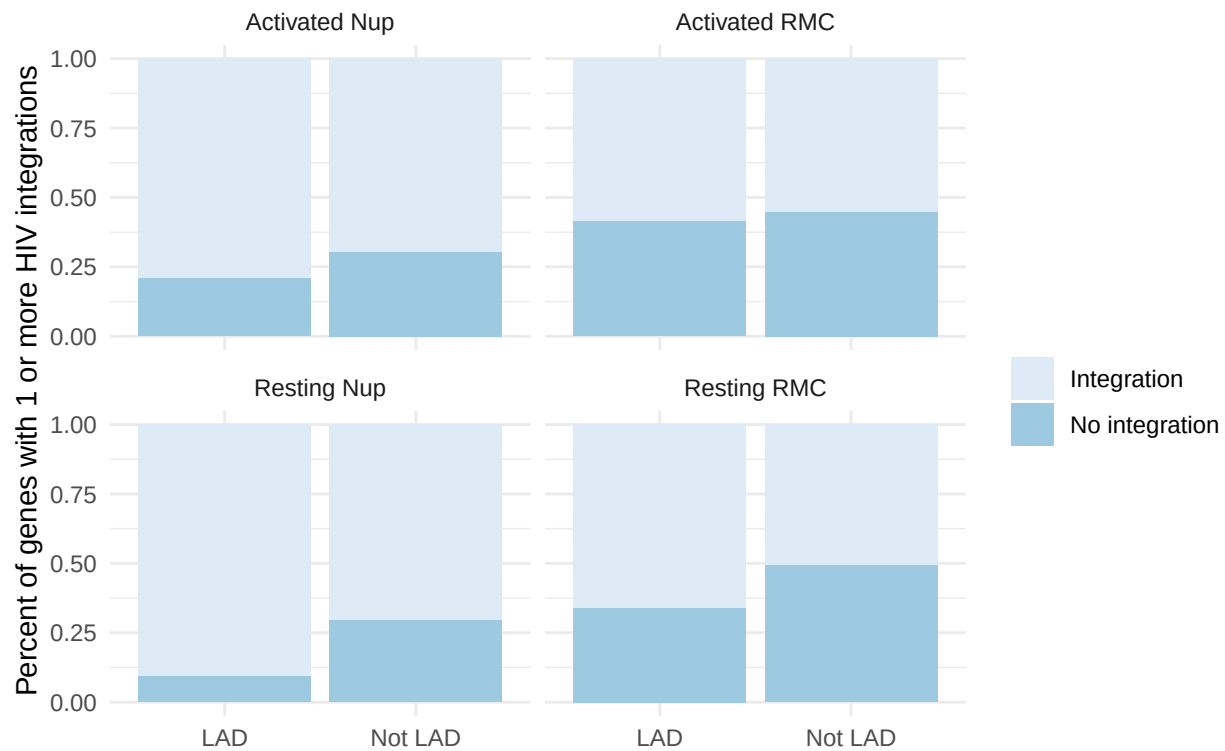
3. Comparison of above with HIV integrations

More integrations in Nup genes than random gene sample. Also Robson LADs overlapping genes seem to have small increase in KO153 group integrations.

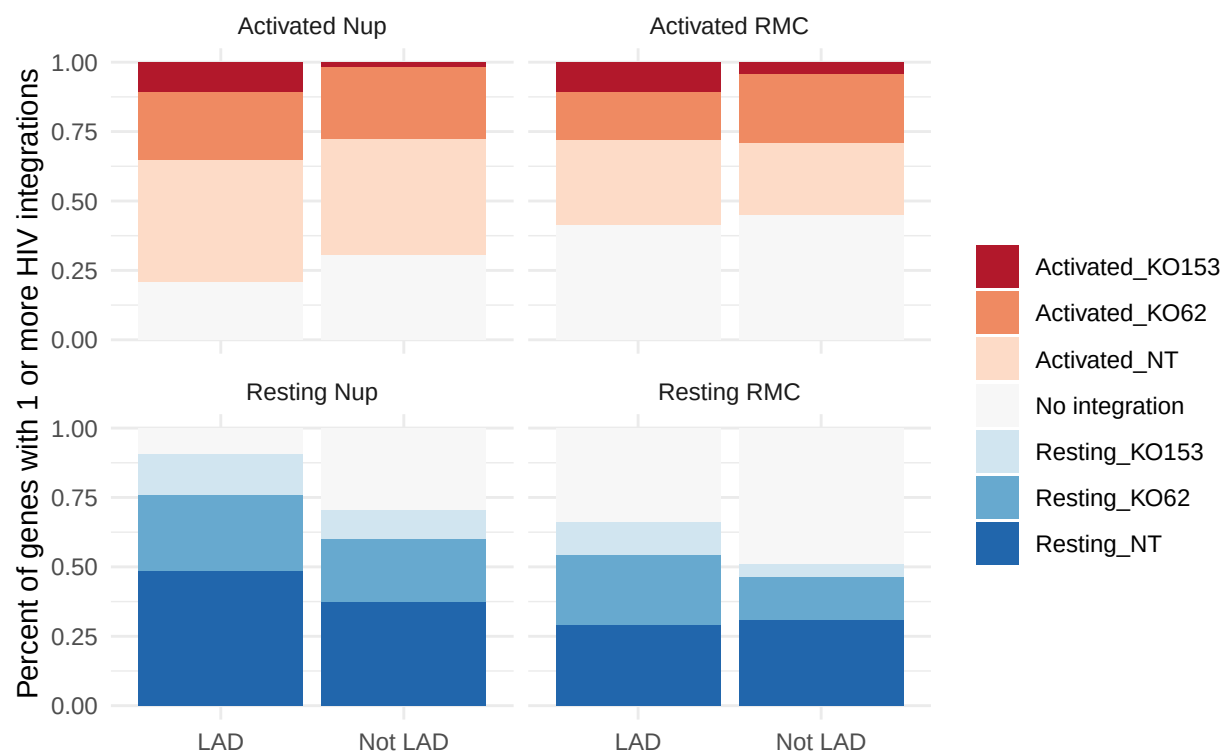
The recurrent integrations graph isn't adjusted for different group sizes yet (NT is about the same size as KO62 and KO153 together by number of integrations). This is unlikely to explain all of the variance between them.

No real difference in percentage of exonal integrations overlapping Activated or Resting LADs, but randomly selected genes have integrations fall in more exonal LAD overlaps.

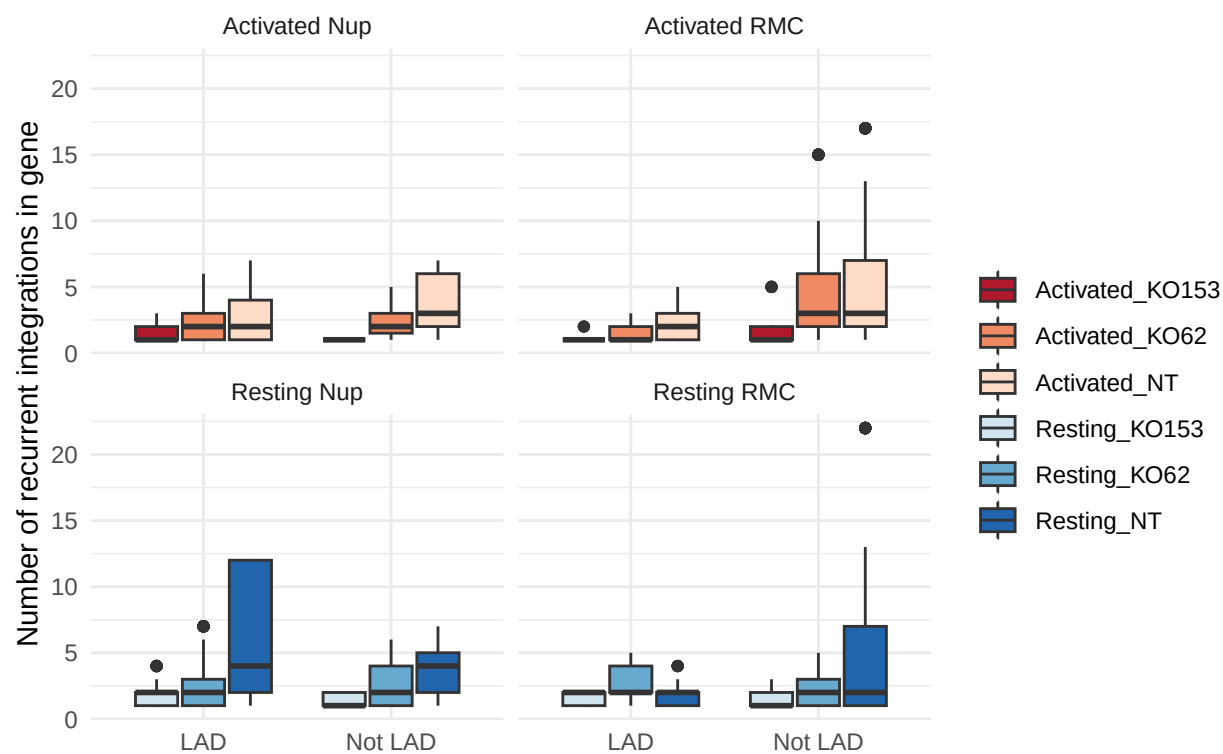
Nup genes with integration split by Robson LAD status



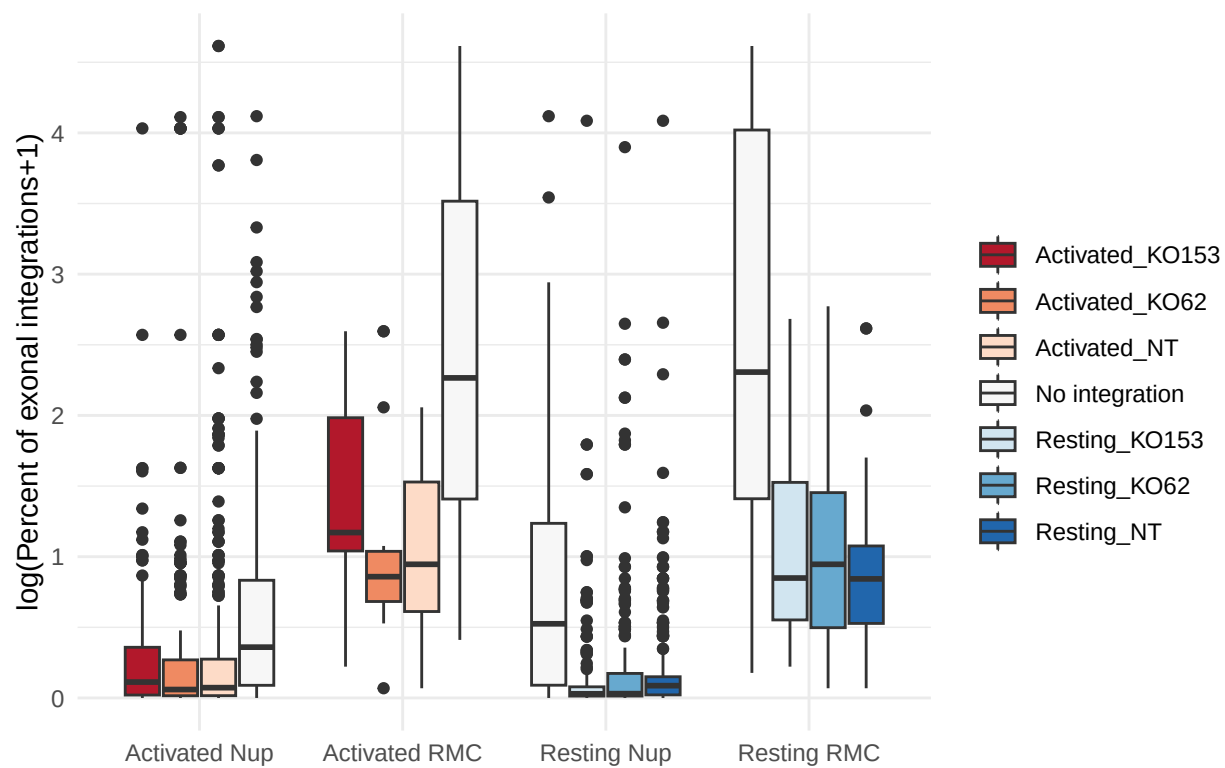
Nup genes with integration split by Robson LAD status



Nup genes with recurrent integration split by Robson LAD status



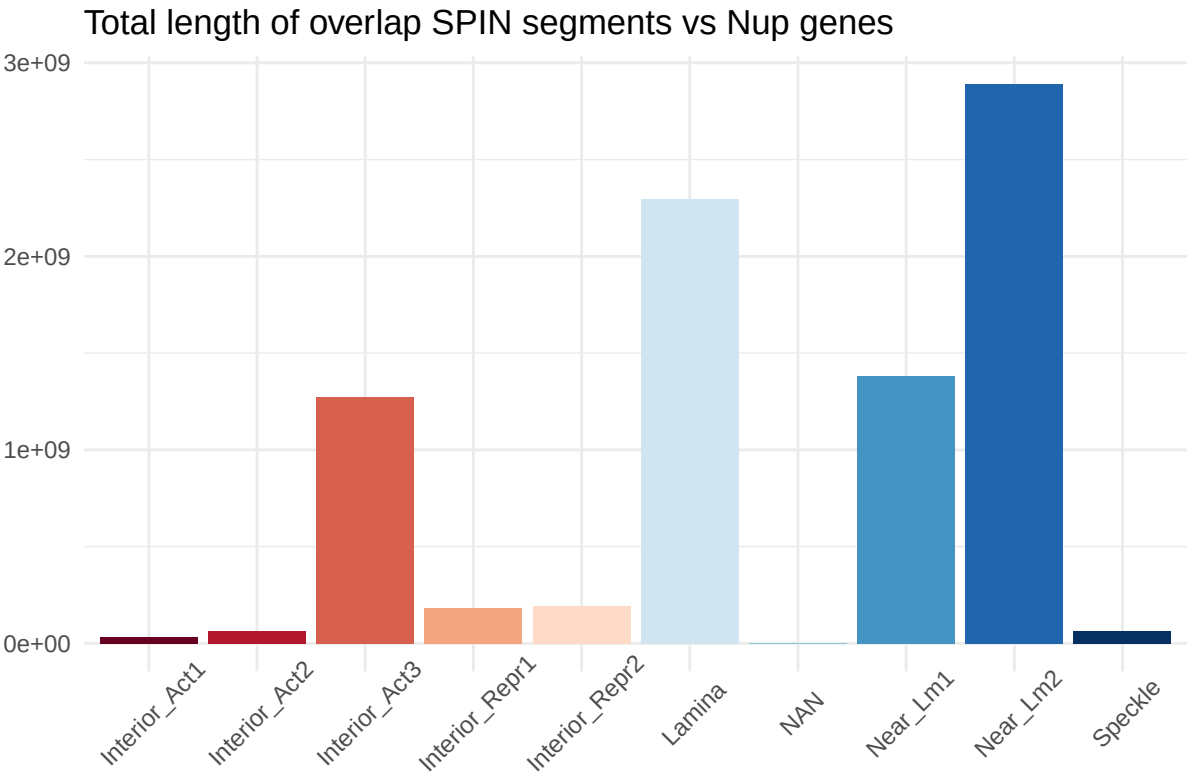
Exonal integration percentage split by Robson LAD status



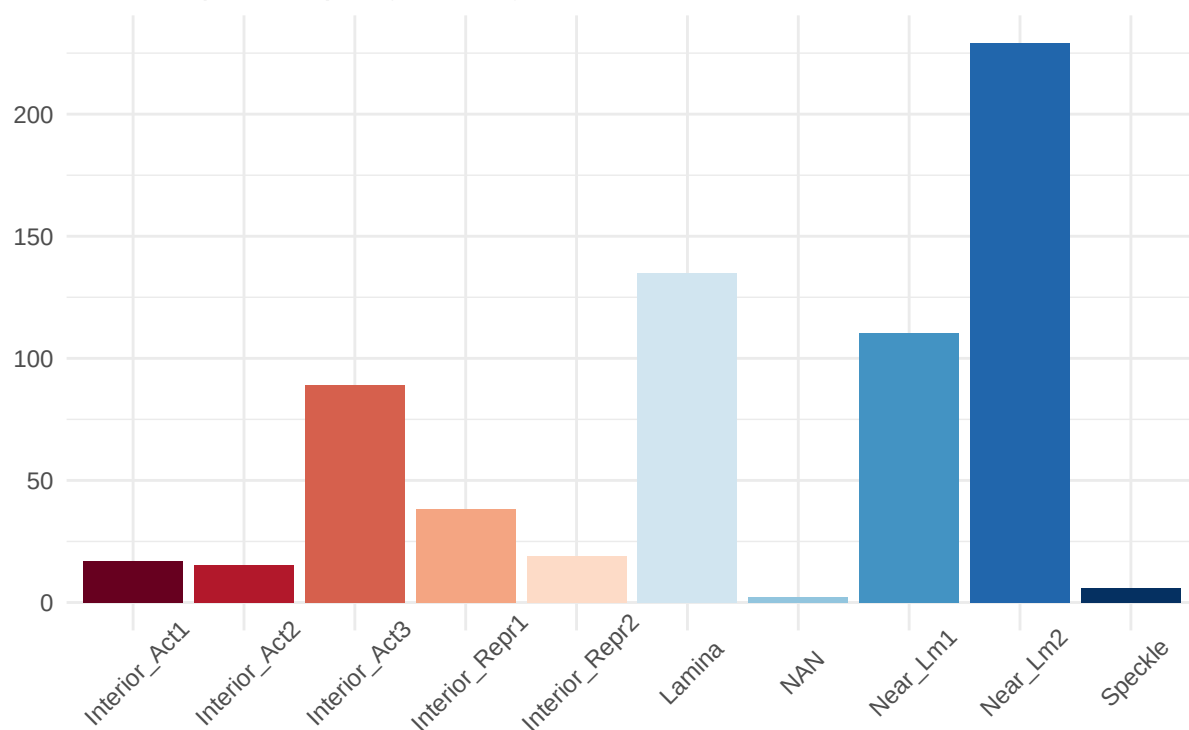
Belmont SPIN clasification vs Nup genes

4 nup genes have no overlap with the Belmont Spin clasification “CISD1” “FAM74A2” “INTS4L2” “ANKRD20A14P”

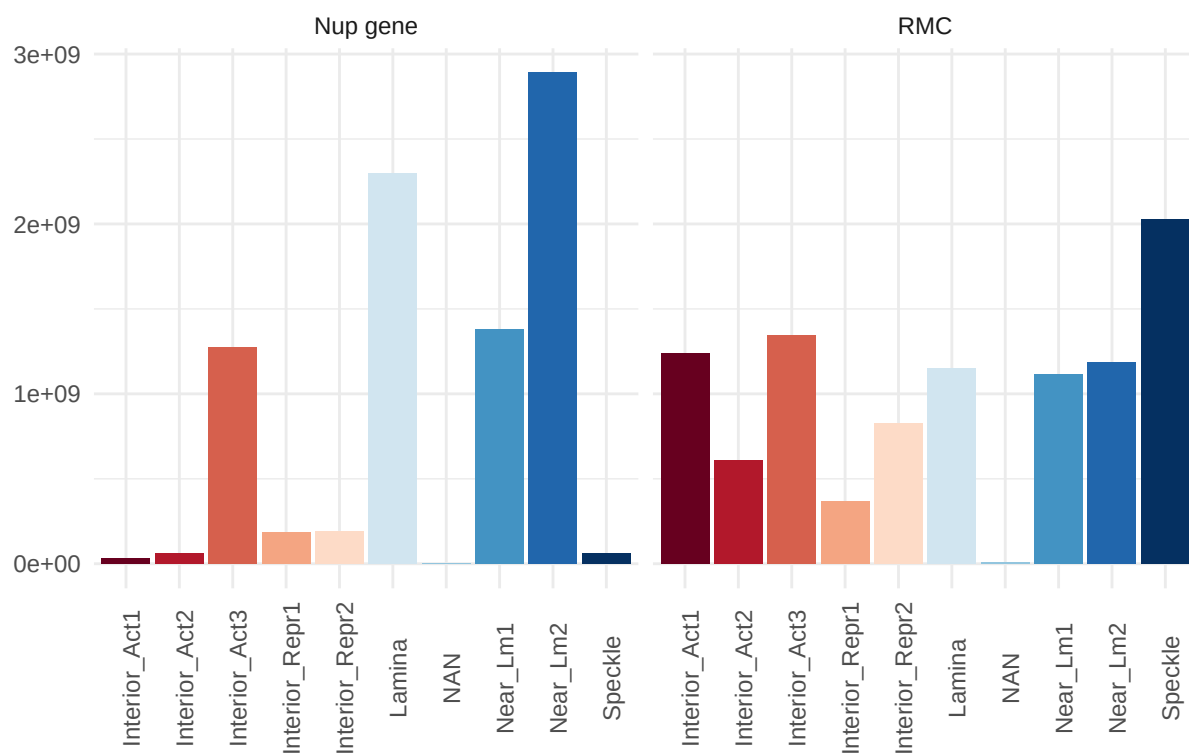
Comparison to randomly selected genes shows avoidance of Speckles and first two interior active compartments.



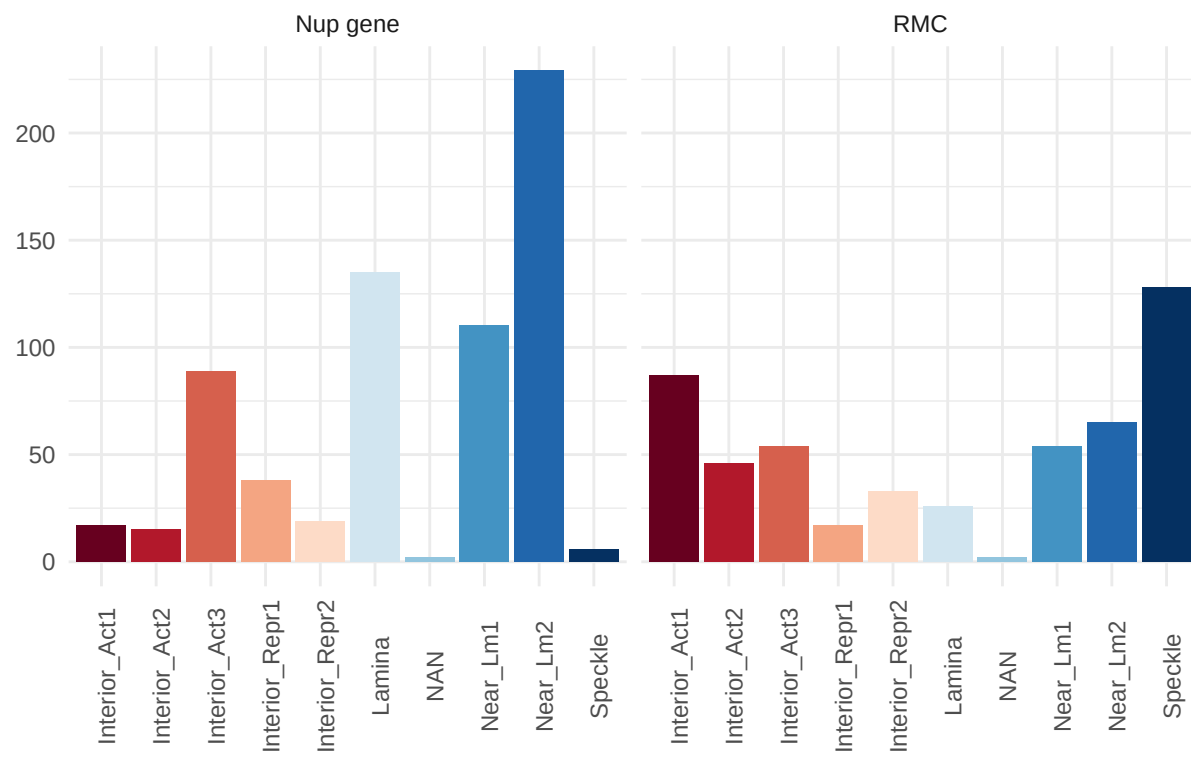
SPIN/Nup overlap adjusted by SPIN class size



Total length of overlap SPIN segments vs Nup genes / RMCs



SPIN/Nup-RMC overlap adjusted by SPIN class size



Checking if theres any additional integration preference here

KO153 remarkable lack of Interior Act 3 compared to others?

Integrations vs Nup gene SPIN classes

